



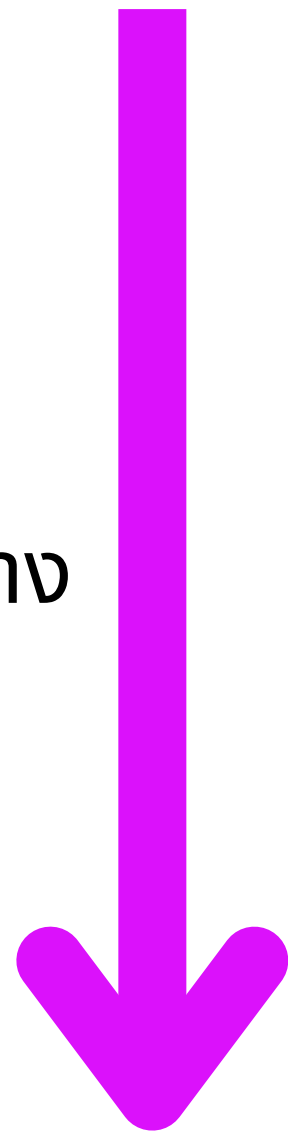
KMUTNB

ภาษา C

เบื้องต้น

การทำงานของโค้ด

ทำงานจาก บน->ล่าง



```
1  #include <stdio.h>
2
3  int a = 5;
4
5  int main(void) {
6      int b = 6;
7      printf("%d\n", a+b);
8      return 0;
9  }
```

Header

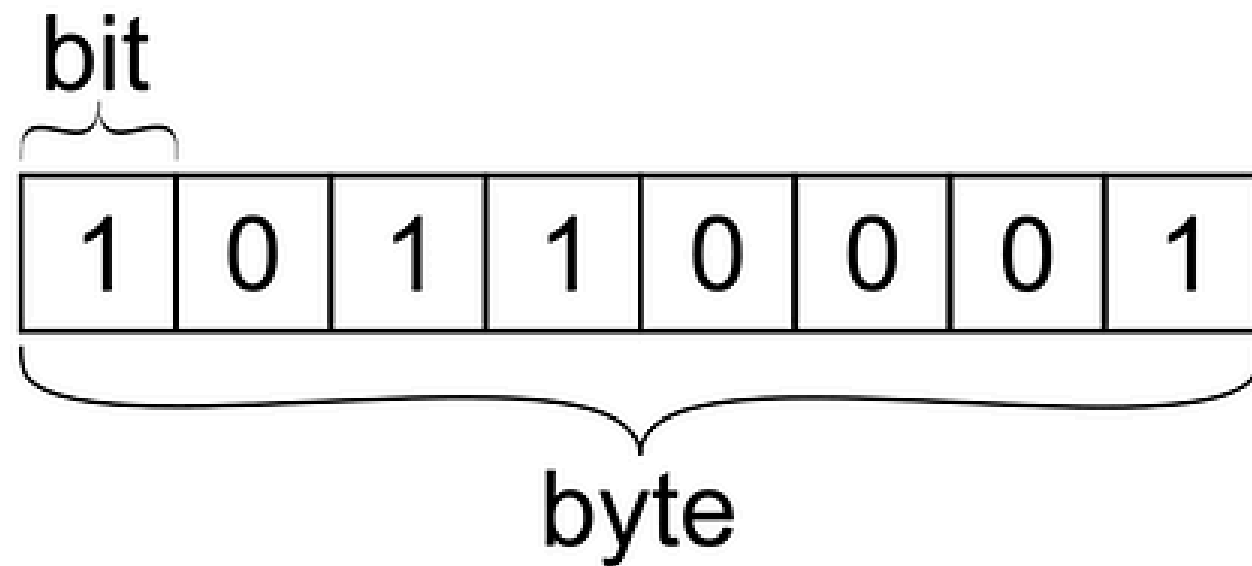
ตัวแปร(Global)

การทำงานหลัก

ตัวแปร

Data type	Size(byte)	Range	Format Specifier
int	4	-2,147,483,648 to 2,147,483,647	%d
unsigned int	4	0 to 4,294,967,295	%u
float	4	1.2E-38 to 3.4E+38	%f
char	1	-128 to 127	%c
unsigned char	1	0 to 255	%c

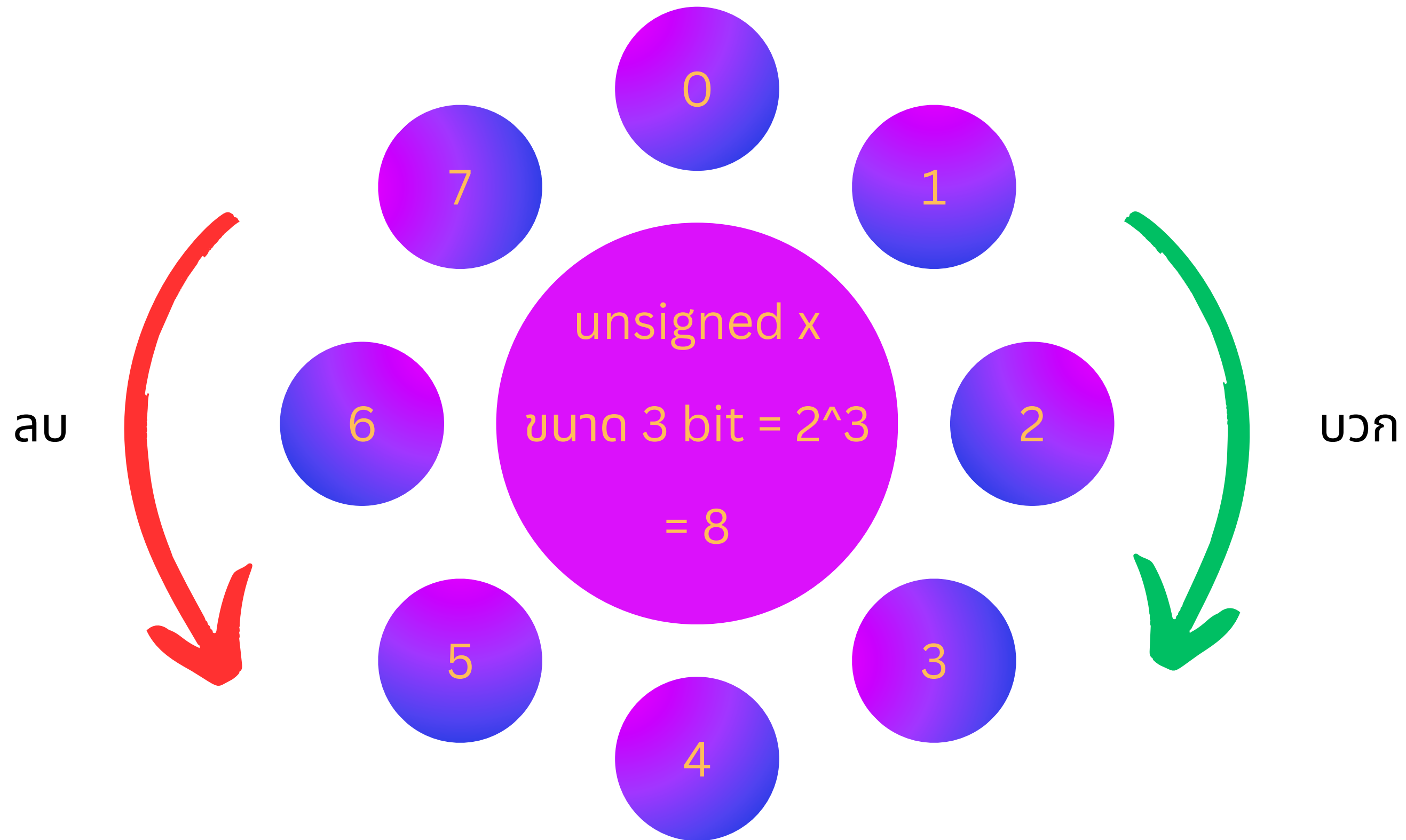
ขนาดของ byte



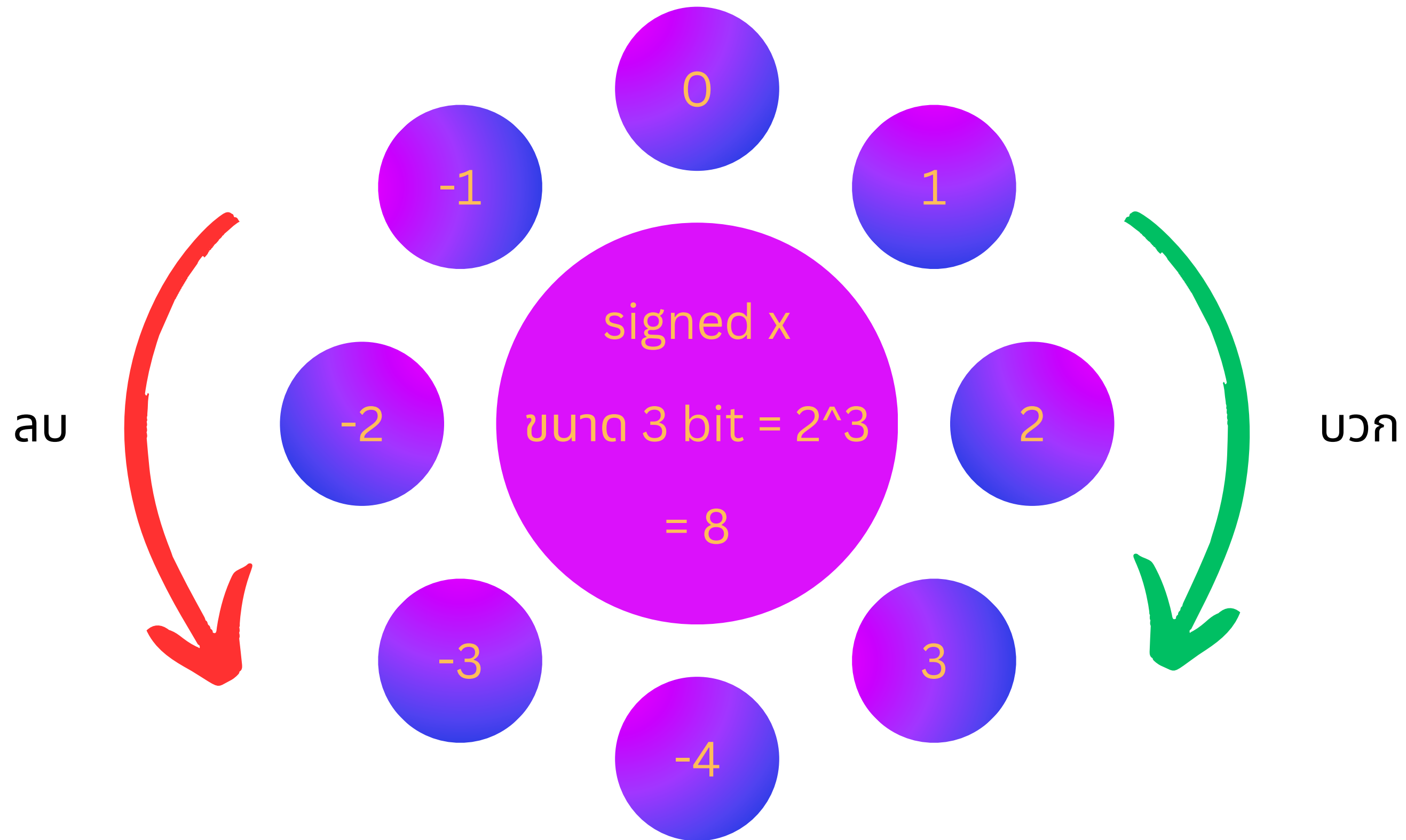
char = 1 byte = 8 bit = $2^8 = 256$

int = 4 byte = 32 bit = $2^{32} = 4,294,967,296$

Range สำคัญยังไง?



Range สำคัญยังไง?



การใช้ printf()

`printf("Hello world!");` → Hello world!

`printf("แอปเปิ้ล %d ลูก", 3);` → แอปเปิ้ล 3 ลูก

`float x = 5.67;`
`printf("Money left : %f bath", x);` → Money left : 5.670000 bath

`float x = 5.67;`
`printf("Money left : %.2f bath", x);` → Money left : 5.67 bath

การใช้ printf()

```
printf("Hello world");  
printf("Hello everyone");
```

```
printf("Hello world\nHello everyone");
```

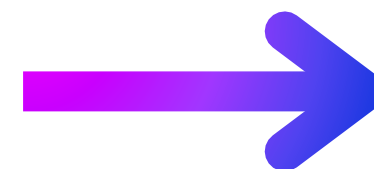
การใช้ printf()

```
printf("Hello world");  
printf("Hello everyone");
```



```
Hello worldHello everyone
```

```
printf("Hello world\nHello everyone");
```



```
Hello world  
Hello everyone
```

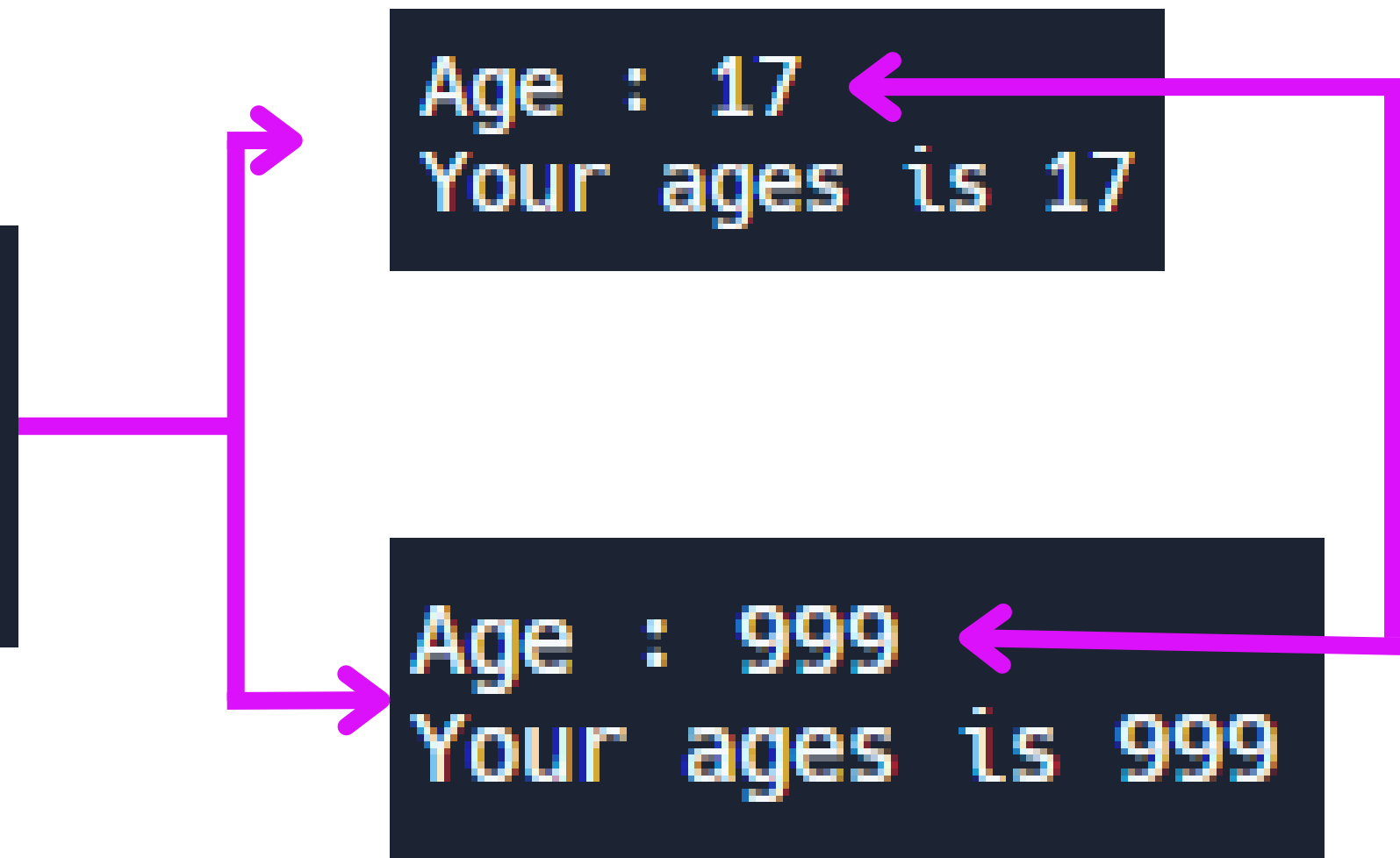

การใช้ scanf()

```
int age;  
printf("Age : ");  
scanf("%d", &age);  
printf("Your ages is %d\n", age);
```

Age : 17
Your ages is 17

Age : 999
Your ages is 999

Input



Arithmetic Operators

Operator	Name	Example
+	Addition	$x + y$
-	Subtraction	$x - y$
*	Multiplication	$x * y$
/	Division	x / y
%	Modulus	$x \% y$
++	Increment	$x++$, $++x$
--	Decrement	$x--$, $--x$

Arithmetic Operators

```
int x = 1;  
++x;  
printf("++x = %d\n", x);
```

```
int x = 1;  
x++;  
printf("x++ = %d\n", x);
```


Arithmetic Operators

```
int x = 1;  
++x;  
printf("++x = %d\n", x);
```



```
++x = 2
```

```
int x = 1;  
x++;  
printf("x++ = %d\n", x);
```



```
x++ = 2
```

Arithmetic Operators

```
int x = 1;  
printf("x++ = %d\n", x++);
```

```
int x = 1;  
printf("++x = %d\n", ++x);
```

Arithmetic Operators

```
int x = 1;  
printf("x++ = %d\n", x++);
```



```
x++ = 1
```

```
int x = 1;  
printf("++x = %d\n", ++x);
```



```
++x = 2
```

Arithmetic Operators

Operators	Example	Same as
<code>+=</code>	<code>x += 2</code>	<code>x = x+2</code>
<code>-=</code>	<code>x -= 3</code>	<code>x = x-3</code>
<code>*=</code>	<code>x *= 4</code>	<code>x = x*4</code>
<code>/=</code>	<code>x /= 5</code>	<code>x = x/5</code>
<code>%=</code>	<code>x %= 6</code>	<code>x = x%6</code>

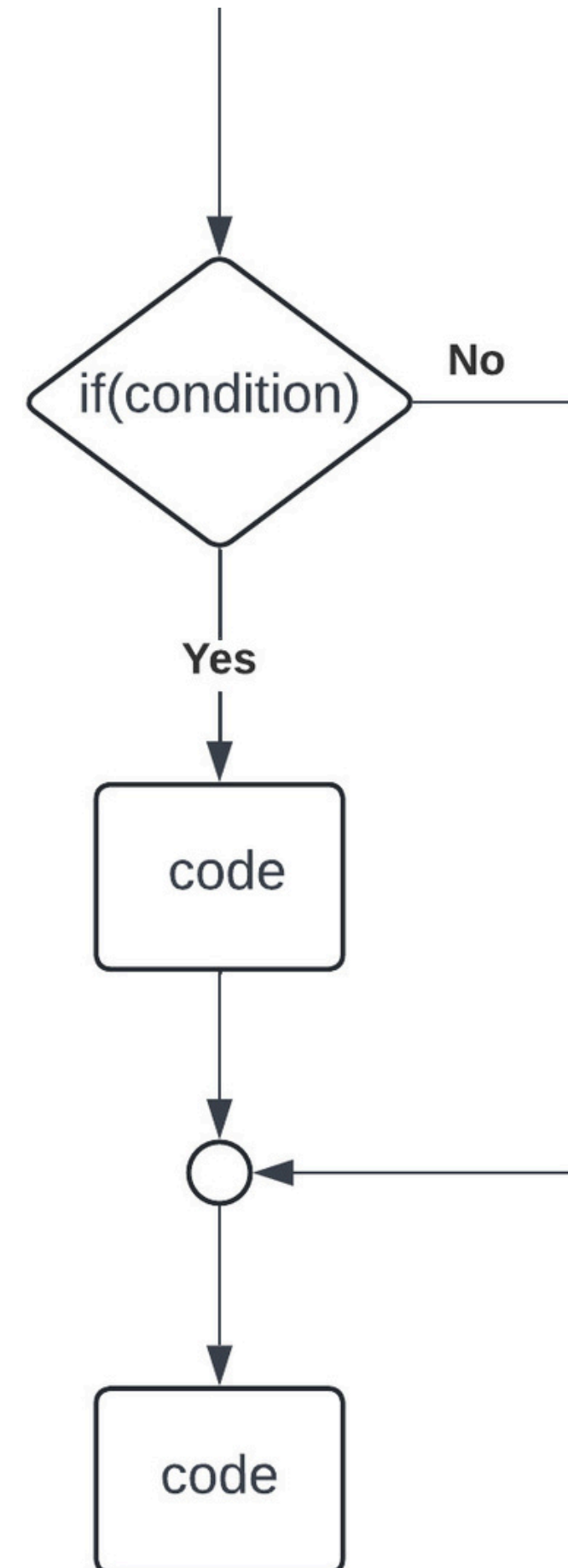
Comparison Operators

Operator	Name	Example
==	Equal	x == y
!=	Not equal	x != y
>	Greater than	x > y
<	Less than	x < y
>=	Greater than or equal	x >= y
<=	Less than or equal	x <= y

Logical Operators

Operator	Name	Example
&&	And	(x>5) && (x<10)
	Or	(x>=8) (y<5)
!	Not	!(5>10)

การทำงานของ if



if

```
int x = 1;  
if(x < 10){  
    printf("Hello\n");  
}  
printf("My name is Rookie\n");
```



```
Hello  
My name is Rookie
```

```
int x = 19;  
if(x < 10){  
    printf("Hello\n");  
}  
printf("My name is Rookie\n");
```



```
My name is Rookie
```


if

```
int x = 5;  
if(x < 10)  
printf("Hello\n");  
printf("My name is Rookie\n");
```



```
Hello  
My name is Rookie
```

```
int x = 19;  
if(x < 10)  
printf("Hello\n");  
printf("My name is Rookie\n");
```



```
My name is Rookie
```

if

```
int x = 5;  
if(x)  
printf("Hello\n");  
printf("My name is Rookie\n");
```

```
int x = 0;  
if(x)  
printf("Hello\n");  
printf("My name is Rookie\n");
```

```
int x = -6;  
if(x)  
printf("Hello\n");  
printf("My name is Rookie\n");
```

if

```
int x = 5;  
if(x)  
printf("Hello\n");  
printf("My name is Rookie\n");
```



```
Hello  
My name is Rookie
```

```
int x = 0;  
if(x)  
printf("Hello\n");  
printf("My name is Rookie\n");
```



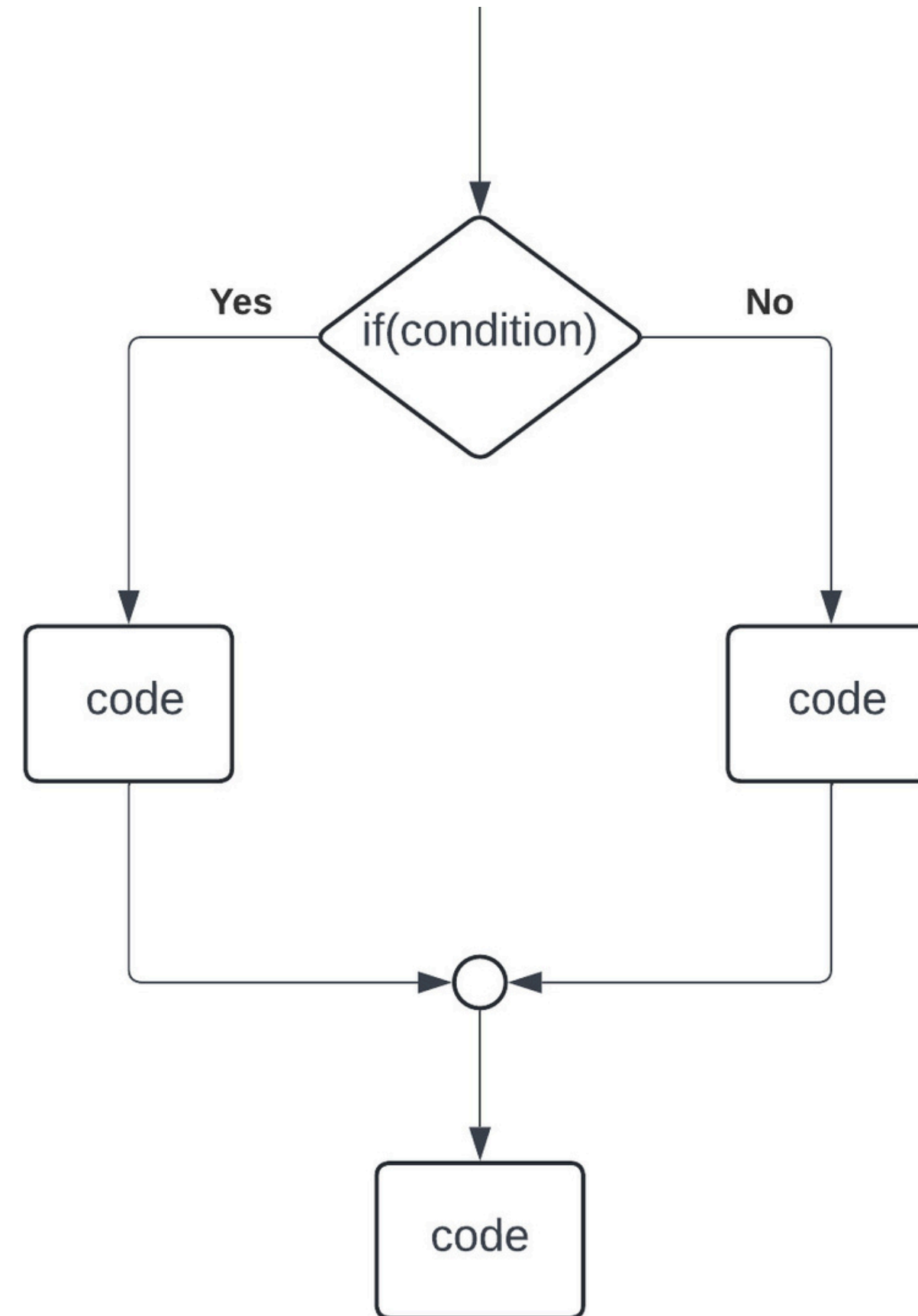
```
My name is Rookie
```

```
int x = -6;  
if(x)  
printf("Hello\n");  
printf("My name is Rookie\n");
```



```
Hello  
My name is Rookie
```

การทำงานของ if...else



if...else

```
int score = 50;  
if(score >= 80){  
    printf("Grade : A\n");  
}  
else{  
    printf("Grade : F\n");  
}
```



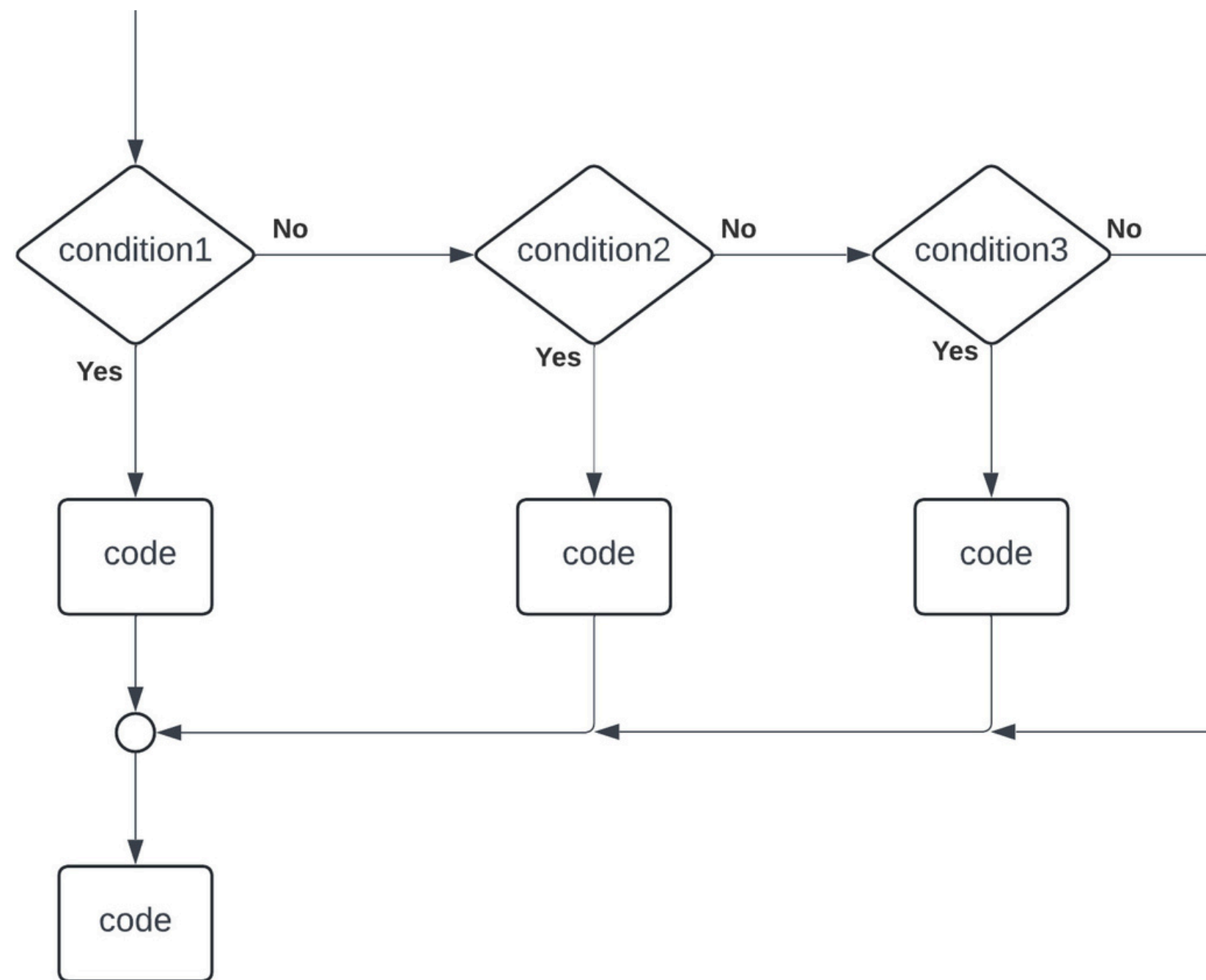
Grade : F

```
int score = 85;  
if(score >= 80){  
    printf("Grade : A\n");  
}  
else{  
    printf("Grade : F\n");  
}
```



Grade : A

การทำงานของ if...else และ else if



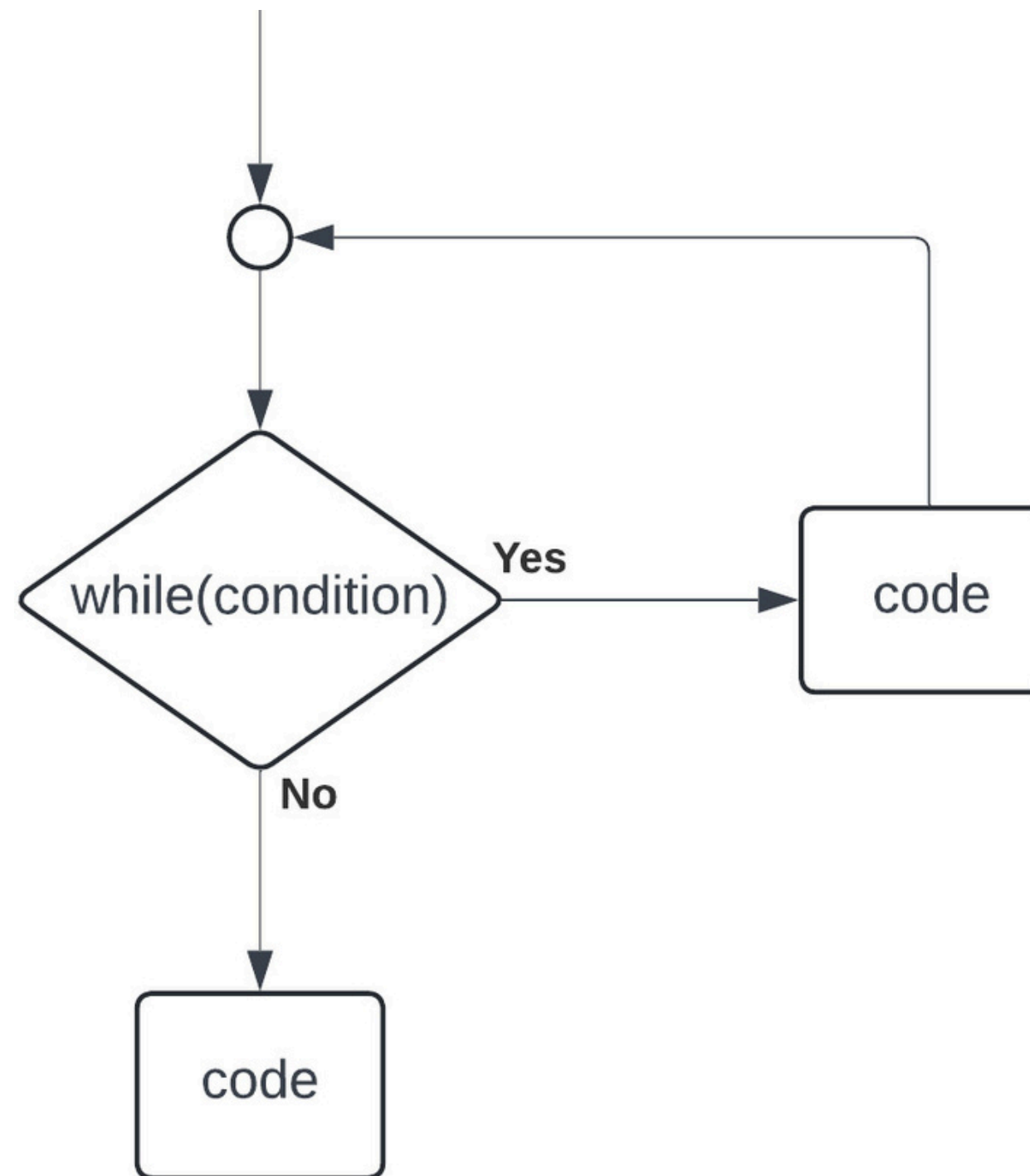
if...else and else if

```
int score = 55;
if(score >= 80){
    printf("Grade : A\n");
}
else if(70 <= score && score < 80){
    printf("Grade : B\n");
}
else if(60 <= score && score < 70){
    printf("Grade : C\n");
}
else if(50 <= score && score < 60){
    printf("Grade : D\n");
}
else{
    printf("Grade : F\n");
}
```

switch...case

```
int day = 4;
switch(day){
    case 1:
        printf("Monday\n");
        break;
    case 2:
        printf("Tuesday\n");
        break;
    case 3:
        printf("Wednesday\n");
        break;
    case 4:
        printf("Thursday\n");
        break;
    case 5:
        printf("Friday\n");
        break;
    case 6:
        printf("Saturday\n");
        break;
    case 7:
        printf("Sunday\n");
        break;
    default:
        printf("Data error\n");
}
```


การทำงานของ while



while

```
int i = 0;
while(i < 5){
    printf("Rookie %d\n", i);
    i++;
}
printf("KMUTNB\n");
```



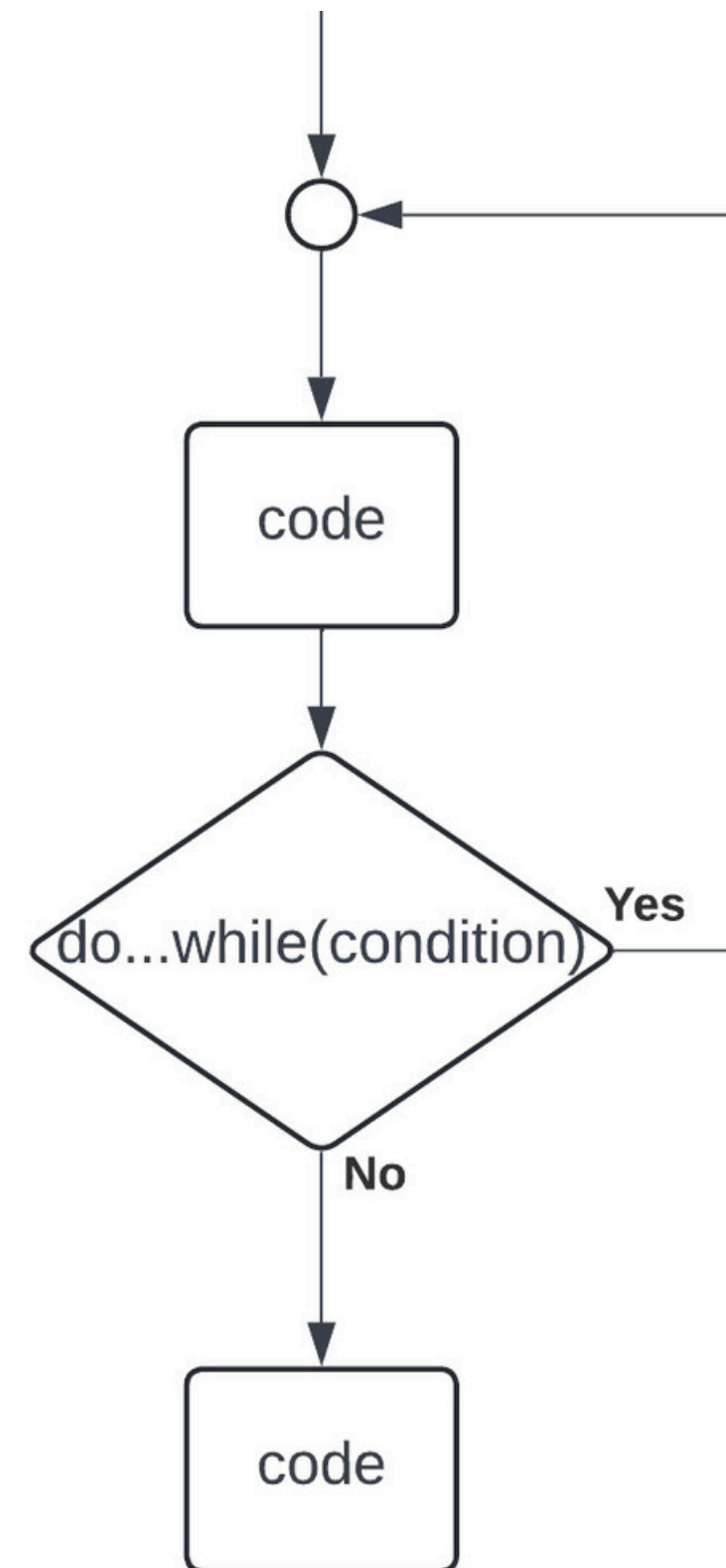
```
Rookie 0
Rookie 1
Rookie 2
Rookie 3
Rookie 4
KMUTNB
```

while

```
int i = 0;
while(i < 5){
    printf("Rookie %d\n", i);
}
printf("KMUTNB\n");
```

[illegible]

การทำงานของ do...while



do...while vs while

```
int i = 0;
do{
    printf("Tata %d\n", i);
    i++;
}
while(i > 0 && i < 5);
printf("KMUTNB\n");
```



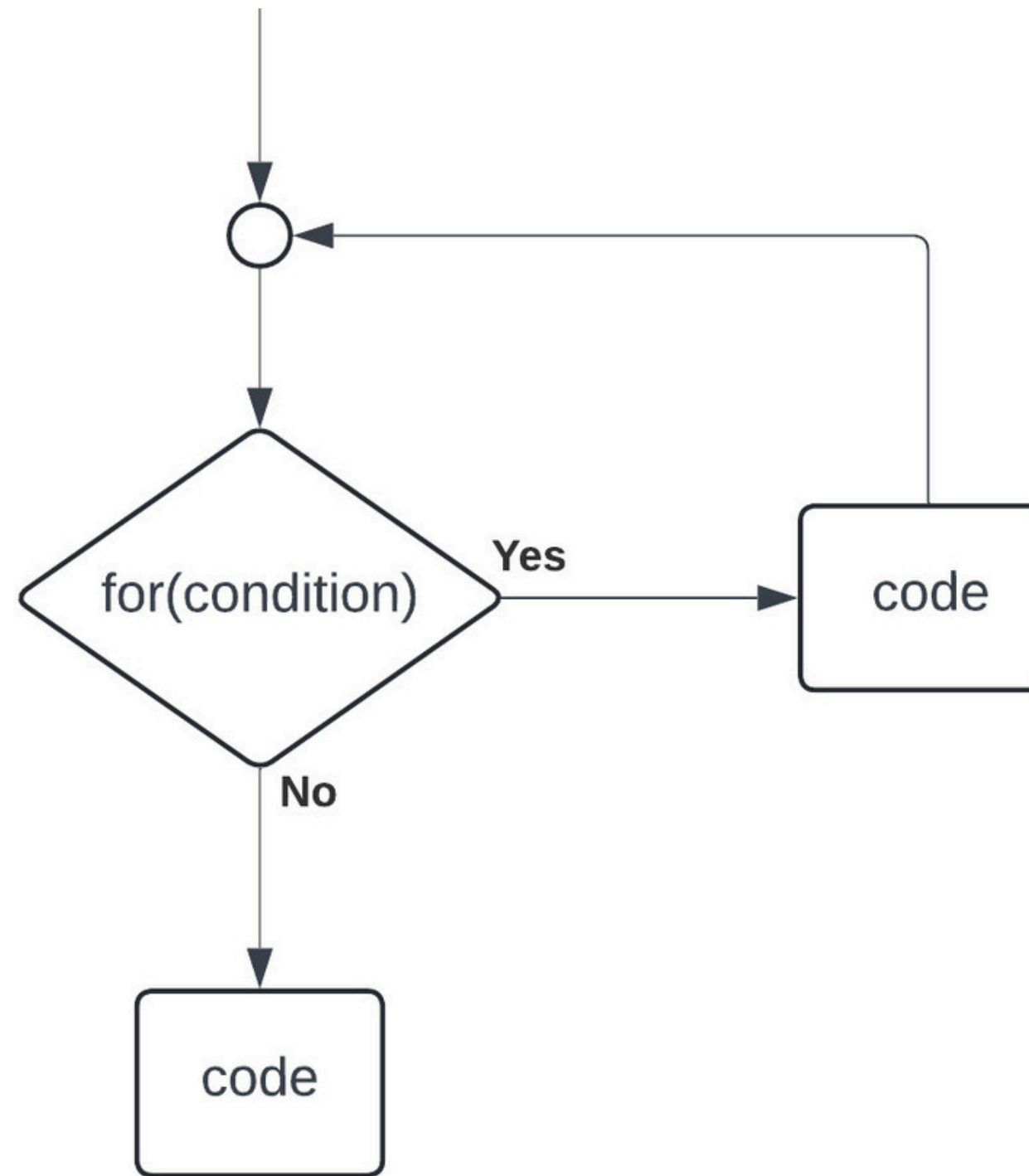
```
Tata 0
Tata 1
Tata 2
Tata 3
Tata 4
KMUTNB
```

```
int i = 0;
while(i > 0 && i < 5){
    printf("Tata %d\n", i);
    i++;
}
printf("KMUTNB\n");
```



```
KMUTNB
```


การทำงานของ for



for vs while

```
for(int i = 0; i < 5; i++){  
    printf("i = %d\n", i);  
}  
printf("KMUTNB\n");
```



```
i = 0  
i = 1  
i = 2  
i = 3  
i = 4  
KMUTNB
```

```
int i = 0;  
while(i < 5){  
    printf("i = %d\n", i);  
    i++;  
}  
printf("KMUTNB\n");
```



```
i = 0  
i = 1  
i = 2  
i = 3  
i = 4  
KMUTNB
```



break/continue

break

หยุดการทำงานของ Loop ทันที

continue

ข้ามไป Loop ถัดไปทันที



break / continue

```
for(int i = 0; i < 5; i++){  
    printf("Bipoe ");  
    if(i == 3){  
        printf("\n");  
        break;  
    }  
    printf("%d\n", i);  
}  
printf("KMUTNB\n");
```



```
Bipoe 0  
Bipoe 1  
Bipoe 2  
Bipoe  
KMUTNB
```

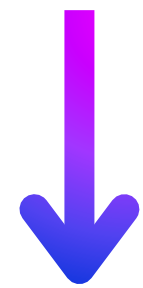
```
for(int i = 0; i < 5; i++){  
    printf("Bipoe ");  
    if(i == 3){  
        printf("\n");  
        continue;  
    }  
    printf("%d\n", i);  
}  
printf("KMUTNB\n");
```



```
Bipoe 0  
Bipoe 1  
Bipoe 2  
Bipoe  
Bipoe 4  
KMUTNB
```

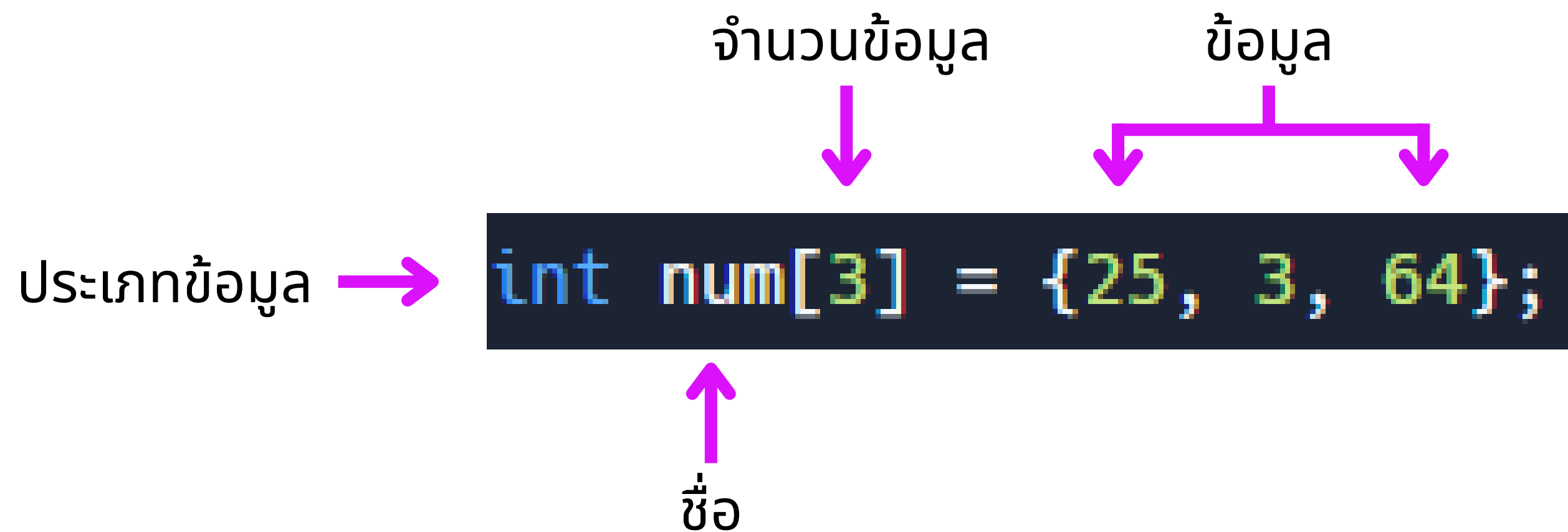
Array

```
int num0 = 25;  
int num1 = 3;  
int num2 = 64;
```



```
int num[3] = {25, 3, 64};
```

การประกาศ Array



```
int num[] = {25, 3, 64};
```

```
int num[3];
```

การใช้งาน Array

```
int num0 = 25;  
int num1 = 3;  
int num2 = 64;  
printf("num0 = %d\n", num0);  
printf("num1 = %d\n", num1);  
printf("num2 = %d\n", num2);  
num0 = 55;  
printf("num0 = %d\n", num0);
```



```
num0 = 25  
num1 = 3  
num2 = 64  
num0 = 55
```

```
int num[3] = {25, 3, 64};  
printf("num0 = %d\n", num[0]);  
printf("num1 = %d\n", num[1]);  
printf("num2 = %d\n", num[2]);  
num[0] = 55;  
printf("num0 = %d\n", num[0]);
```



```
num0 = 25  
num1 = 3  
num2 = 64  
num0 = 55
```

ประโยชน์ของ Array

```
int num0 = 25;  
int num1 = 3;  
int num2 = 64;  
int num3 = 44;  
int num4 = 21;  
printf("num0 = %d\n", num0);  
printf("num1 = %d\n", num1);  
printf("num2 = %d\n", num2);  
printf("num3 = %d\n", num3);  
printf("num4 = %d\n", num4);
```



```
num0 = 25  
num1 = 3  
num2 = 64  
num3 = 44  
num4 = 21
```

```
int num[5] = {25, 3, 64, 44, 21};  
for(int i = 0; i < 5; i++){  
    printf("num%d = %d\n", i, num[i]);  
}
```



```
num0 = 25  
num1 = 3  
num2 = 64  
num3 = 44  
num4 = 21
```


ประโยชน์ของ Array

```
int num0 = 25;
int num1 = 3;
int num2 = 64;
int num3 = 44;
int num4 = 21;

int sum = num0 + num1 + num2 + num3 + num4;
printf("sum = %d\n", sum);
```



SUM = 157

```
int num[5] = {25, 3, 64, 44, 21};

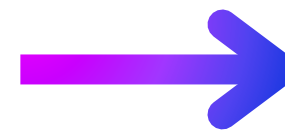
int sum = 0;
for(int i = 0; i < 5; i++){
    sum += num[i];
}
printf("sum = %d\n", sum);
```



SUM = 157

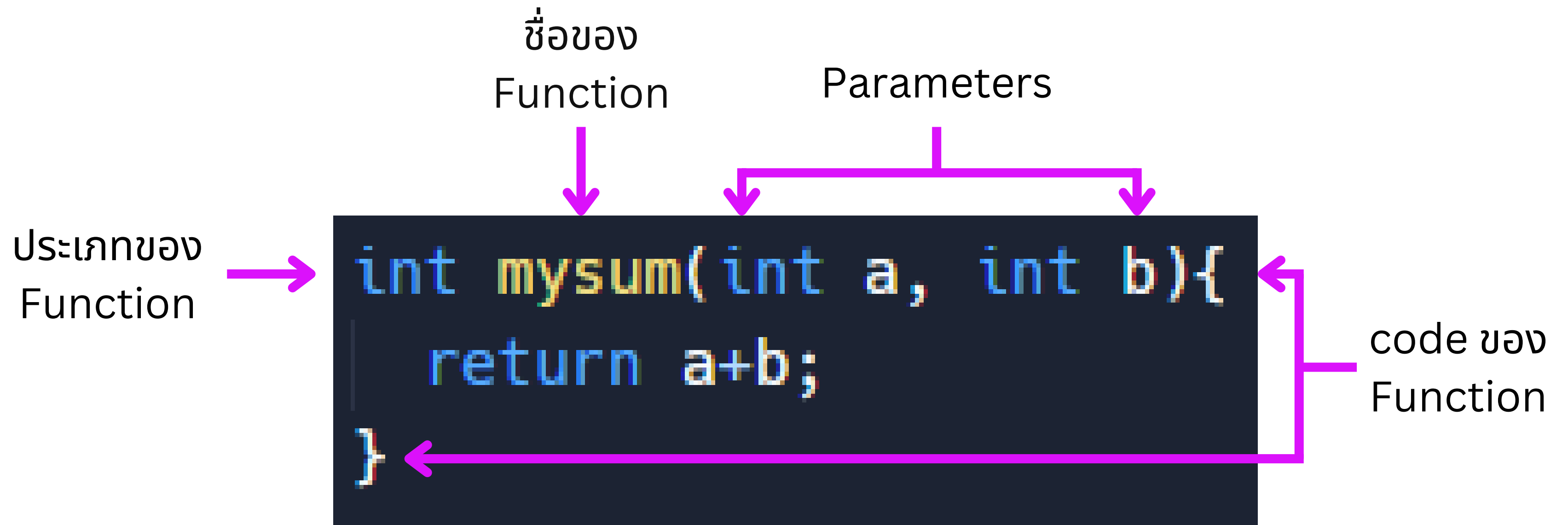
Function

```
int main(void) {  
    printf("Hello World!\n");  
    printf("Hello World!\n");  
    printf("Hello World!\n");  
    return 0;  
}
```



```
void hello(){  
    printf("Hello World!\n");  
    return;  
}  
  
int main(void) {  
    hello();  
    hello();  
    hello();  
    return 0;  
}
```

Function



ตัวอย่างการใช้ Function

```
int main(void) {  
    int a = 3;  
    int b = 8;  
    int c = 16;  
  
    int ab = a + b;  
    int ac = a + c;  
    int bc = b + c;  
    return 0;  
}
```



```
int mysum(int a, int b){  
    return a+b;  
}  
  
int main(void) {  
    int a = 3;  
    int b = 8;  
    int c = 16;  
  
    int ab = mysum(a, b);  
    int ac = mysum(a, c);  
    int bc = mysum(b, c);  
    return 0;  
}
```

ข้อควรระวัง Function

```
void change(int a){  
    a = 8;  
}  
  
int main(void) {  
  
    int a = 3;  
    printf("a = %d\n", a);  
    change(a);  
    printf("a = %d\n", a);  
  
    return 0;  
}
```



```
a = 3  
a = 3
```

ແລນ

```
void change(int* a){  
    *a = 8;  
}  
  
int main(void) {  
    int a = 3;  
    printf("a = %d\n", a);  
    change(&a);  
    printf("a = %d\n", a);  
    return 0;  
}
```



a	=	3
a	=	8



KMUTNB

นี่เป็นแค่เบื้องต้น

ขอให้โชคดีสำหรับการเรียน